

CloudFront Cache Invalidation for Updated Content

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon S3** static website was used as the origin for an **Amazon CloudFront** distribution so the effects of edge caching could be observed. The S3 bucket was intentionally created in a region far from the viewer to make cache behavior easier to notice. After the website was updated in the origin, the CloudFront URL continued showing the older content because the edge location still served cached data. To refresh the content immediately, a CloudFront invalidation was created for the affected path. Finally, the updated page appeared correctly after the invalidation completed, confirming that cache invalidation forces CloudFront to fetch fresh content from the origin.

Understanding CloudFront Invalidation

Amazon CloudFront stores copies of website content at edge locations so users can load pages faster from nearby servers. When the origin content changes, CloudFront may continue serving the older cached version until the cache expires or is manually cleared. Invalidation exists so you can tell CloudFront to remove specific cached files or all cached objects immediately. This is useful when a website update must appear right away instead of waiting for the TTL to end.

Example:

A homepage in `index.html` is changed from “Welcome” to “New Update” in the S3 origin. Users opening the S3 website endpoint see the new text immediately, but users opening the CloudFront URL still see the old version until an invalidation is created for `/*`. After invalidation, CloudFront fetches the fresh file and displays the updated page.

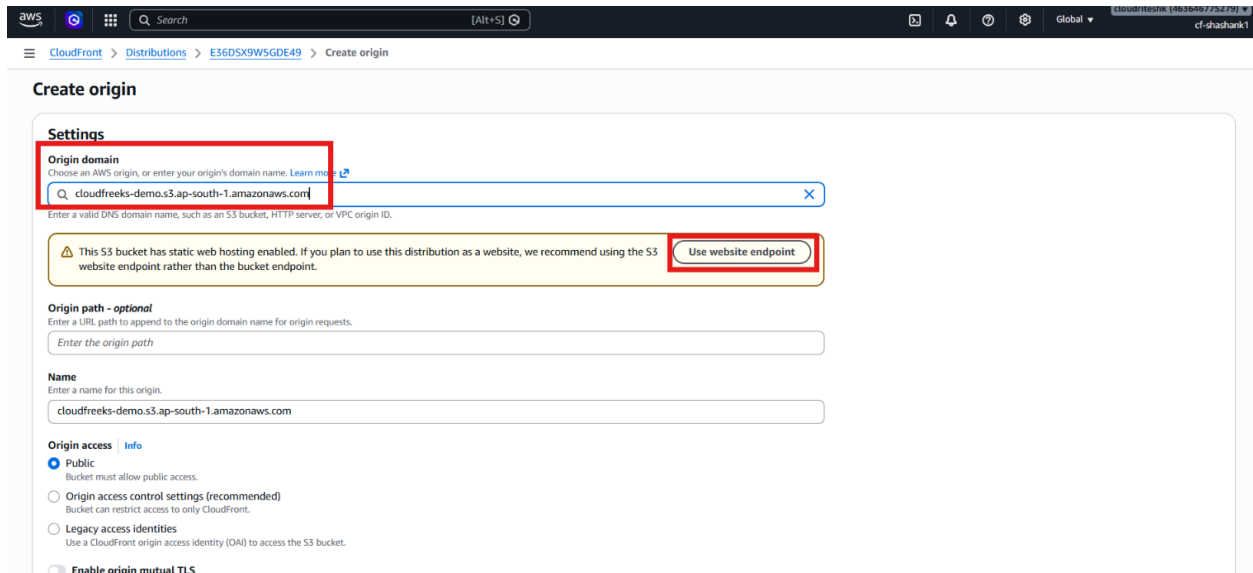
Lab Steps

Step 1 — Create the S3 Static Website Origin

- Sign in to the AWS Management Console.
- Open **Amazon S3**.
- Create or select the static website bucket in a region away from your location.
- Enable static website hosting.
- Upload the website files, including `index.html`.
- Verify that the S3 website endpoint opens correctly.

Step 2 — Create the CloudFront Distribution

- Open **Amazon CloudFront**.
- Click **Create distribution**.
- Select the S3 bucket as the origin.
- Choose the website endpoint when prompted.



Create origin

Settings

Origin domain
Choose an AWS origin, or enter your origin's domain name. [Learn more](#)

cloudfreeks-demo.s3.ap-south-1.amazonaws.com

Enter a valid DNS domain name, such as an S3 bucket, HTTP server, or VPC origin ID.

This S3 bucket has static web hosting enabled. If you plan to use this distribution as a website, we recommend using the S3 website endpoint rather than the bucket endpoint. **Use website endpoint**

Origin path - optional
Enter a URL path to append to the origin domain name for origin requests.
Enter the origin path

Name
Enter a name for this origin.
cloudfreeks-demo.s3.ap-south-1.amazonaws.com

Origin access [Info](#)

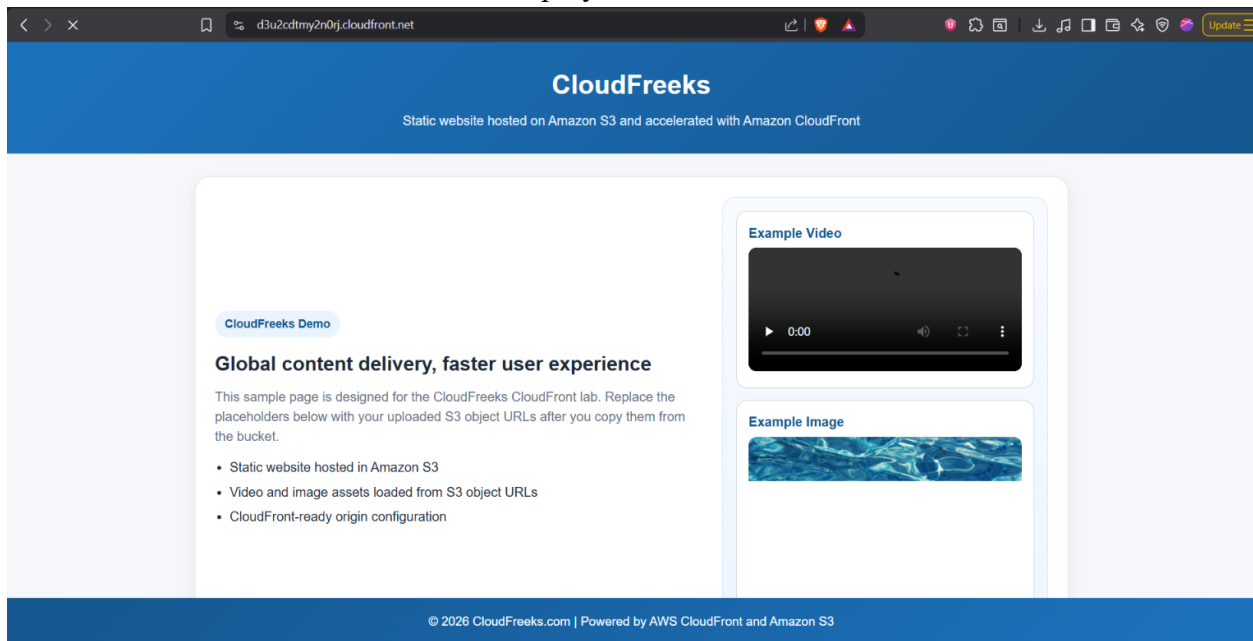
☒ **Public**
Bucket must allow public access.

☐ **Origin access control settings (recommended)**
Bucket can restrict access to only CloudFront.

☐ **Legacy access identities**
Use a CloudFront origin access identity (OAI) to access the S3 bucket.

☐ **Enable origin mutual TLS**

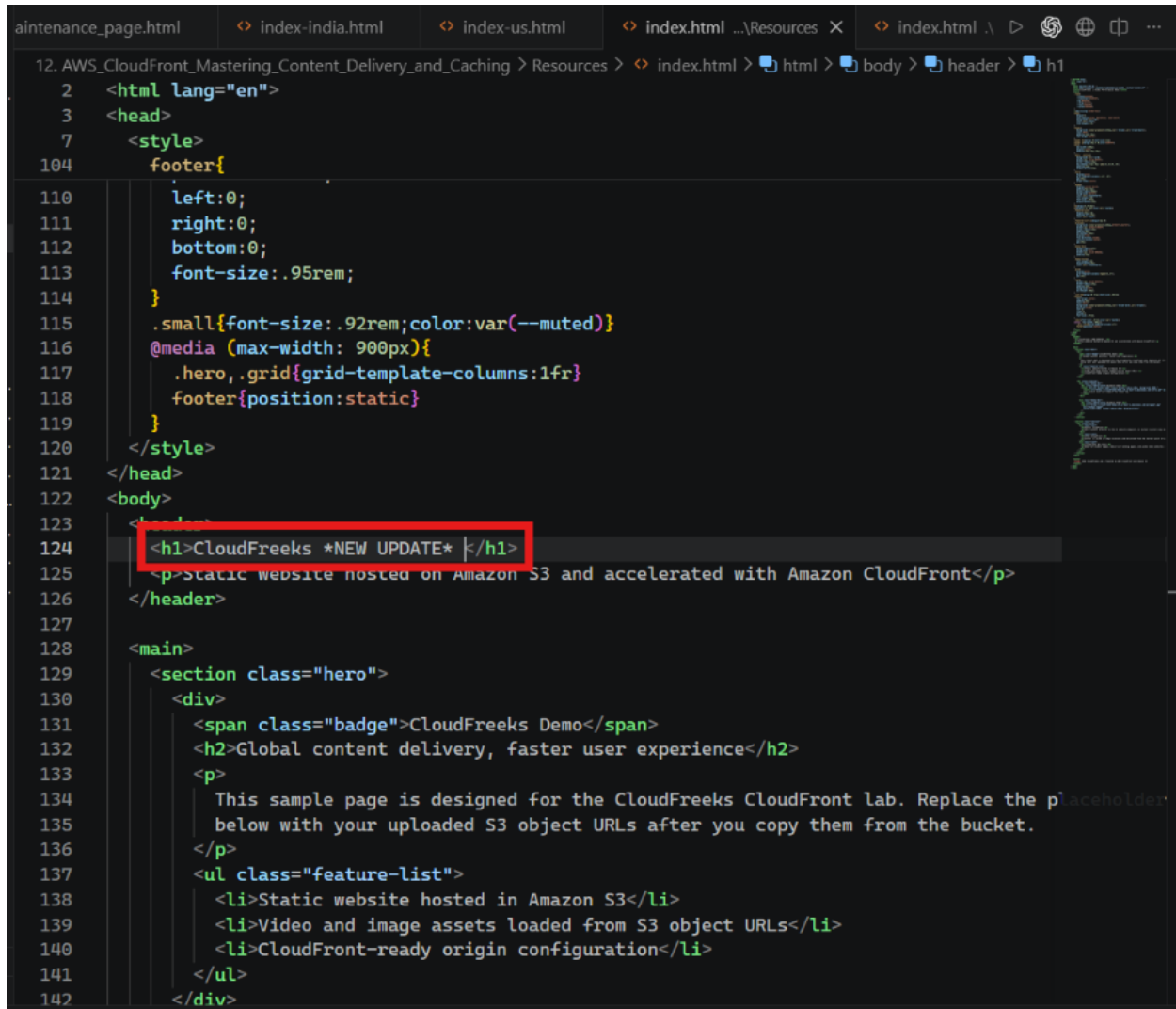
- Leave the remaining settings as default.
- Disable **Web Application Firewall** for the lab.
- Create the distribution and wait for deployment.



- Verify that the website opens through the CloudFront URL.

Step 3 — Update the Origin Content

- Open the website source file in a code editor.

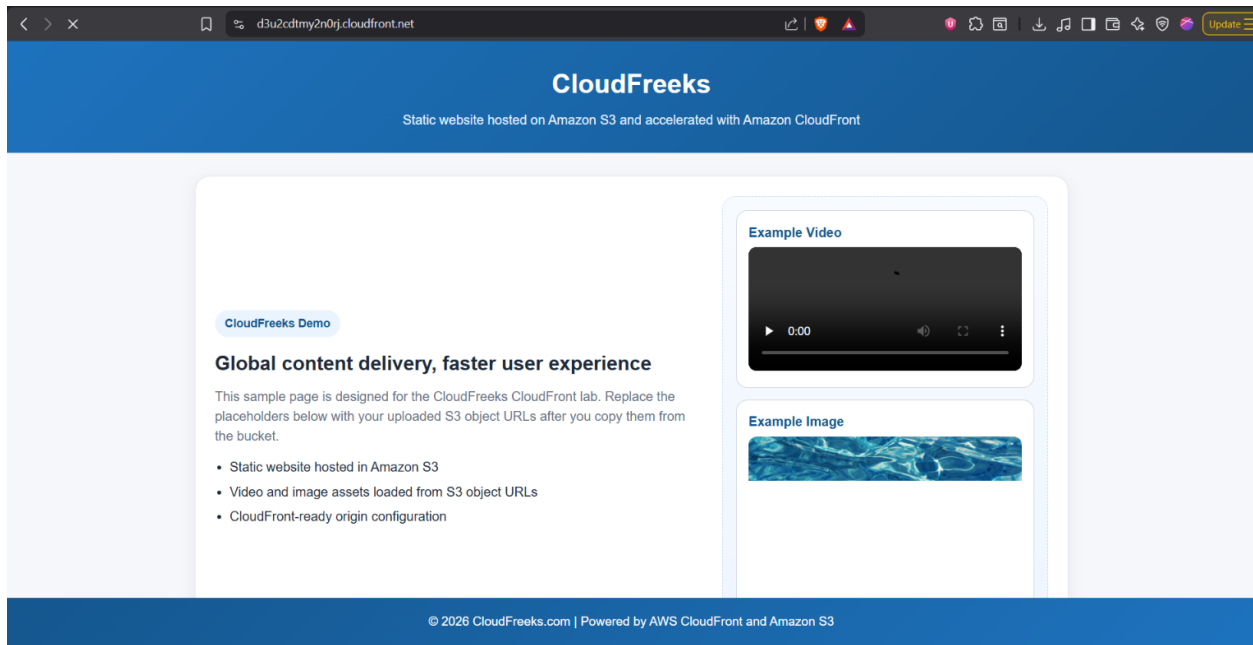


```
12. AWS_CloudFront_Mastering_Content_Delivery_and_Caching > Resources > index.html > html > body > header > h1
 2  <html lang="en">
 3  <head>
 7    <style>
104    footer{
110      left:0;
111      right:0;
112      bottom:0;
113      font-size:.95rem;
114    }
115    .small{font-size:.92rem;color:var(--muted)}
116    @media (max-width: 900px){
117      .hero,.grid{grid-template-columns:1fr}
118      footer{position:static}
119    }
120  </style>
121 </head>
122 <body>
123   <header>
124     <h1>CloudFreeks *NEW UPDATE* </h1>
125     <p>Static website hosted on Amazon S3 and accelerated with Amazon CloudFront</p>
126   </header>
127
128   <main>
129     <section class="hero">
130       <div>
131         <span class="badge">CloudFreeks Demo</span>
132         <h2>Global content delivery, faster user experience</h2>
133         <p>
134           This sample page is designed for the CloudFreeks CloudFront lab. Replace the placeholder
135           below with your uploaded S3 object URLs after you copy them from the bucket.
136         </p>
137         <ul class="feature-list">
138           <li>Static website hosted in Amazon S3</li>
139           <li>Video and image assets loaded from S3 object URLs</li>
140           <li>CloudFront-ready origin configuration</li>
141         </ul>
142       </div>
```

- Modify index.html by adding a new update to the main heading.
- Save the file.
- Upload the updated index.html back to the S3 bucket.
- Verify that the S3 website endpoint shows the updated content.

Step 4 — Observe Cached Content in CloudFront

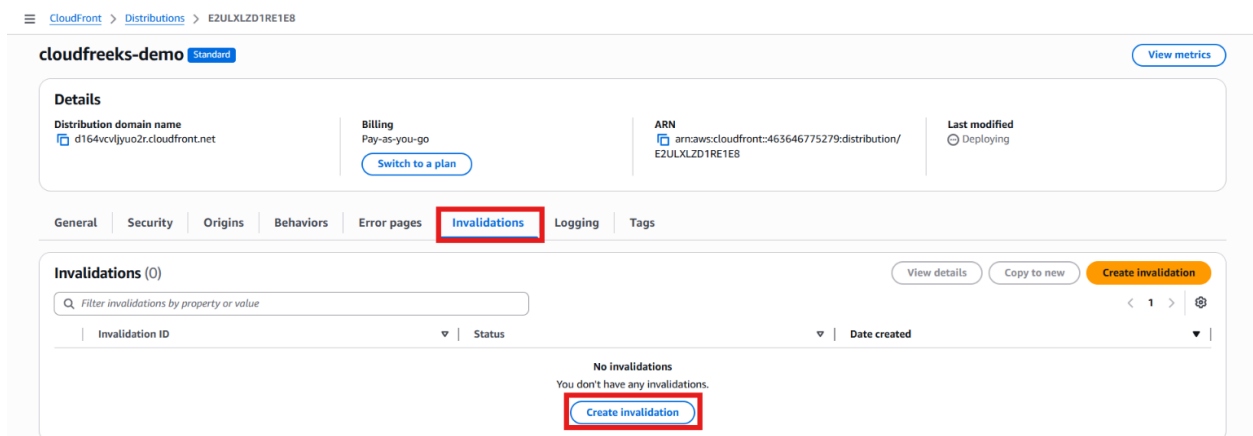
- Open the CloudFront URL in a browser.
- Refresh the page and confirm that the old content is still being served.



- Open the site in an incognito window to avoid browser cache.
- Verify that the update is not visible yet through CloudFront.

Step 5 — Create an Invalidation and Verify the Update

- Open the CloudFront distribution.
- Navigate to the **Invalidations** tab.
- Click **Create invalidation**.



- Enter the invalidation path: /*

Create invalidation
Choose which cached resources will be invalidated so that endpoints will load them again.

Object paths
Selection method

☒ By paths
Specify the paths of the objects to invalidate. You may use wildcards.

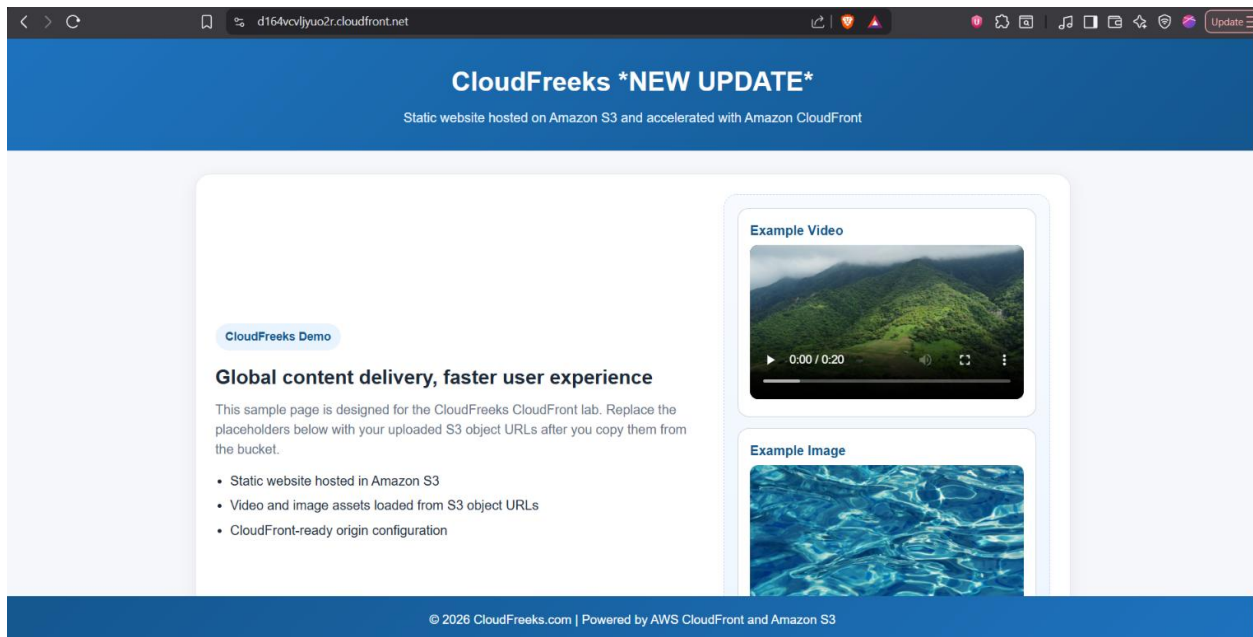
☐ By cache tags
Specify objects to invalidate based on the cache tags set in an HTTP header.

Object paths to invalidate Info
Put each path to invalidate on a separate line. Wildcards (*) are only valid at the end of the path.

/

Cancel **Create invalidation**

- Submit the invalidation request.
- Wait for the status to change to **Completed**.
- Refresh the CloudFront URL.



- Verify that the updated content now appears.
- Open the URL in an incognito window and confirm the same updated result.
- Verify that CloudFront cache invalidation successfully forced the edge location to fetch the new version from the S3 origin.